Tutorial 6

Indian Institute of Technology Jodhpur
Mathematics-I (MAL1010)
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1. Evaluate the following limits.

   (a) $\lim\limits_{x \to 0} \dfrac{\sin(5x)}{x}$

   (b) $\lim\limits_{x \to \infty} \dfrac{3x^3 + 2x + 1}{x^3 - x^2 + 5}$

   (c) $\lim\limits_{x \to 2} \dfrac{x^2 - 4}{x - 2}$

   (d) $\lim\limits_{x \to 0} \dfrac{1 - \cos(x)}{x^2}$

2. Let $f(x)$ be defined as

$$f(x) = \begin{cases} x^2 \sin\left(\frac{1}{x}\right) & x \neq 0 \\ 0 & x = 0 \end{cases}$$

   (a) Is $\lim\limits_{x \to 0} f(x)$ equal to $f(0)$?

   (b) Prove the limit $\lim\limits_{x \to 0} f(x)$ using the ε-δ definition.

   (c) Is $f(x)$ continuous at $x = 0$?

   (d) Is $f(x)$ differentiable at $x = 0$?

3. Evaluate the limit using the ε-δ definition:

$$\lim_{x \to 3} (2x + 1) = 7$$

4. Is the following function continuous at $x = 1$?

$$f(x) = \begin{cases} \frac{x^2 - 1}{x - 1} & x \neq 1 \\ 2 & x = 1 \end{cases}$$

5. Find the number of continuous functions on $\mathbb{R}$ that map rational points to irrational and irrational points to rational numbers.

6. Find the limit and check the differentiability of the function f at $x = c$, where

   (a) $f(x) = x[x]$, c $= 0$

   (b) $f(x) = \frac{\sqrt{x+1} - 1}{x}$, c $= 0$

   (c) $f(x) = [x] - [3x + 1]$, c $= 5$

7. A function $f : \mathbb{R} \to \mathbb{R}$ is defined by $f(0) = 0$ and

$$f(x) = \begin{cases} 0, & \text{if } x \text{ is irrational} \\ \frac{1}{q}, & \text{if } x = \frac{p}{q}, \text{ where } p \in \mathbb{Z}, q \in \mathbb{N} \text{ and } \gcd(p, q) = 1 \end{cases}$$

   Show that f is not differentiable at 0.

8. If a function $f : [a, b] \to \mathbb{R}$ be monotone on $[a, b]$, then the set of points of discontinuities of f in $[a, b]$ is countable.

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